Nearly Linear Sparsification of ℓ_p Subspace Approximation

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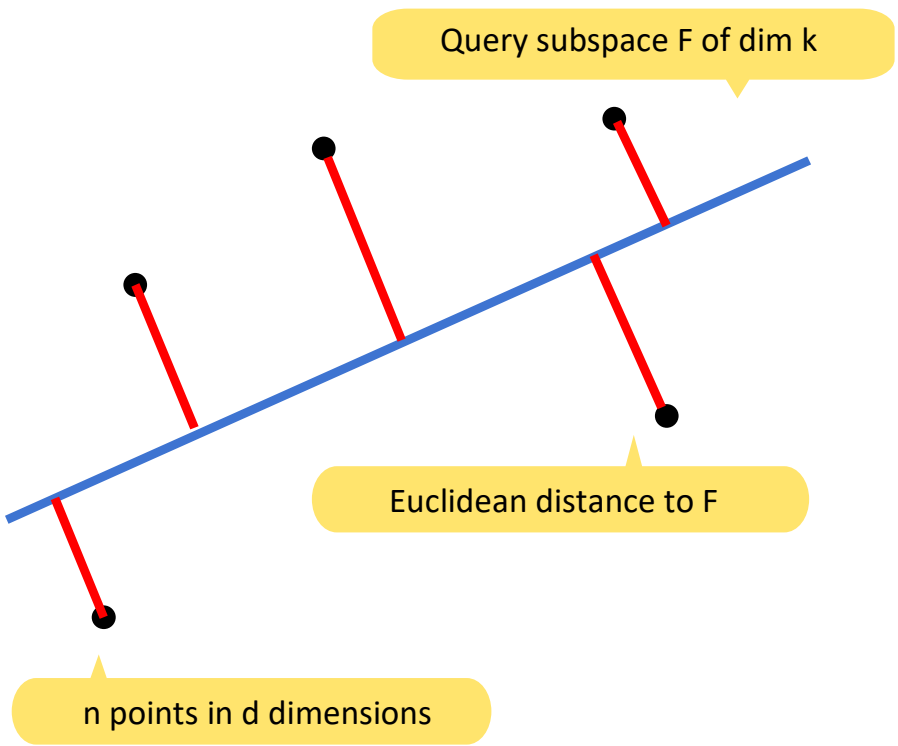
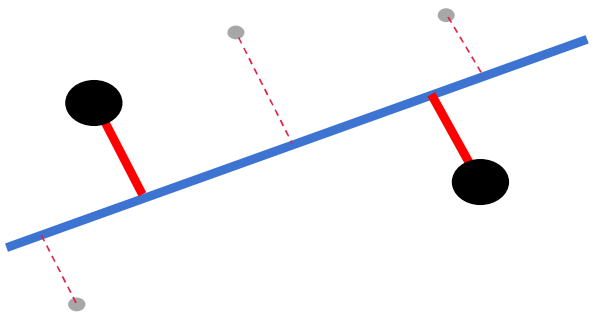
ℓ_p Subspace Approximation



- Cost of a k-dimensional subspace F: sum of p-th powers of the Euclidean distances to F
- Goal: minimize the cost over subspaces F
- p = 2: Principal Component Analyss
- p = 1
  - "Median hyperplane problem"
  - Robust to outliers
  - Rotationally invariant L1 PCA
- p = ∞
  - "Center hyperplane problem"
  - Achieves fairness by minimizing the max distance
  - Generalizes extent/containment problems: enclosing sphere, cylinder

Notation: for $x \in R^d$, its ℓ_p-norm is $\left(\sum_{i=1,\ldots d} |x_i|^p\right)^{1/p}$

Coresets for ℓ_p Subspace Approximation



A : n x d matrix with the n points in the rows

- S : n x n diagonal matrix of weights
- P_F :projection matrix onto subspace F
- $\|\cdot\|_{p,2}$ is (p, 2) norm (ℓ_p norm of ℓ_2 norm of rows)
- OPT is cost of optimal solution in the $\|\cdot\|_{p,2}$ norm

Coreset guarantee:

$$\|SA(I - P_F)\|_{p,2} = (1 \pm \varepsilon)\|A(I - P_F)\|_{p,2}$$

- Exact algorithms are slow [DTV, GRSW, CW] so we need coresets!
- Coreset: weighted subset of the points whose cost approximates the cost of the entire set for every subspace F up to $(1 + \varepsilon)$ factor
- Prior results:
  - [Feldman-Langberg 2011]
    - Coreset of size $\mathrm{poly}(k, d, \varepsilon^{-1})$
  - [Sohler-W 2018]
    - Coreset of size $k^{\max\{1,\frac{p}{2}\}}\ \varepsilon^{-O(p^2)}$
    - Needs an additional coordinate, exponential time
    - [Feng-Kacham-W 2020] fast time, loses $\mathrm{poly}(k, \varepsilon^{-1})$ factors
  - [Huang-Vishnoi 2020]
    - Coreset of size $\mathrm{poly}(k, \varepsilon^{-1})$ for constant p
    - No additional coordinate, fast time
- Main question: best of all worlds?

Coresets for ℓ_p Subspace Approximation

Main Theorem: There is a fast algorithm which constructs a coreset of $k\epsilon^{-4/p}$ points for $1 \leq p < 2$, and $k^{p/2}\epsilon^{-p}$ points for p > 2

- Optimal dependence on k and improved dependence on ϵ for all p. Input sparsity time.
- Surprisingly, we sample based on (root) ridge leverage scores (RLS), which were designed for p = 2
- Key idea to make RLS work for ℓ_p norms: flattening
- Resolves online/streaming settings

Technical Ingredients

Technical Approach: Direct Affine Embedding

- Proof entirely avoids the **Representative Subspace Theorem** [SW18, HV19]

$$A(I - P_F) = A(P^* - P_F) + A(I - P^*)$$

Low Rank AX
(rank ≤ 2k)

Fixed Affine B
(independent of F)

- Fixed component A(I – P*) handled by prior work [Woodruff-Yasuda '23]
- Focus of talk: Preserving the Low Rank component. Running example is p = 1
- Show: $|\mathrm{SAX}|_{\mathrm{p},2} = (1 \pm \epsilon)|\mathrm{AX}|_{\mathrm{p},2}$ for any X with rank ≤ 2k and operator norm ≤ 2

Ridge Leverage Scores

- [Cohen-Musco-Musco 2017] Sampling by ridge leverage scores (RLS) gives nearly optimal coresets for $p = 2$
- Ridge leverage score for i-th row: $a_i^T(A^TA + \lambda I)^{-1}a_i$, for $\lambda = \frac{1}{k}\|A - A_k\|_F^2$
- i-th ridge leverage score upper bounds i-th rank-k sensitivity: $\sup_{\text{rank-}\ P} \frac{\left|a_i^T(I-P)\right|_2^2}{|A(I-P)|_F^2}$. Sum of scores is at most 2k
- Fast algorithms for approximating all ridge leverage scores
- Sample each row independently proportional to its score
- Key property: *additive-multiplicative subspace embedding*

$p = 2$

Lemma. If S is constructed by RLS sampling with $\widetilde{O}(\frac{k}{\varepsilon^2})$ rows, then for all $x \in \mathbb{R}^d$:

$$\|SAx\|_2^2 = \|Ax\|_2^2 \pm \varepsilon(\|Ax\|_2^2 + \lambda\|x\|_2^2)$$

$p = 1$ (ours)

Lemma. If S is constructed by root RLS sampling with $\widetilde{O}(\frac{n^{1/2}k^{1/2}}{\varepsilon^2})$ rows, then for all $x \in \mathbb{R}^d$:

$$\|SAx\|_1 = \|Ax\|_1 \pm \varepsilon(\|Ax\|_1 + \lambda^{1/2}\|x\|_1)$$

Ridge Leverage Scores

Lemma: If S is constructed by root RLS sampling with $\widetilde{O}(\frac{n^{1/2}k^{1/2}}{\varepsilon^2})$ rows, then for all $x \in \mathbb{R}^d$:

$$\|SAx\|_1 = \|Ax\|_1 \pm \varepsilon(\|Ax\|_1 + \lambda^{1/2}\|x\|_1)$$

- Recurse to get O~(k ϵ^{-4}) samples
- Dvoretsky: if G is a Gaussian matrix with enough columns, then $|X|_{1,2} = (1 \pm \epsilon)\ |XG|_{1,1}$ for any X
- $\|SAXG\|_{1,1} = \|AXG\|_{1,1} \pm \varepsilon(\|AXG\|_{1,1} + \lambda^{1/2}\|XG\|_{1,1})$
- Dvoretsky: for any X: $\|SAX\|_{1,2} = \|AX\|_{1,2} \pm \varepsilon(\|AX\|_{1,2} + \lambda^{1/2}\|X\|_{1,2})$
- Recall $\lambda = \frac{1}{k}\|A - A_k\|_F^2$. How to bound the error $\lambda^{1/2}\|X\|_{1,2}$?

Problems with Additive Error

- For any X: $\|SAX\|_{1,2} = \|AX\|_{1,2} \pm \varepsilon(\|AX\|_{1,2} + \lambda^{1/2}\|X\|_{1,2})$, where $\lambda = \frac{1}{k}\|A - A_k\|_F^2$
- Can show $\|X\|_{1,2} \leq n^{\frac{1}{2}}\|X\|_{2,2} \leq k^{\frac{1}{2}} \cdot n^{1/2}$ since X is a rank-k matrix with operator norm at most 1
- Additive error is: $\lambda^{1/2}\|X\|_{1,2} \quad = \frac{1}{k^{1/2}}\|A - A_k\|_F \|X\|_{1,2} \quad \leq \|A - A_k\|_F \cdot n^{\frac{1}{2}} \leq OPT \cdot n^{1/2}$
- This is off by $n^{1/2}$
- Bounding $\|A - A_k\|_F \leq OPT$ is too loose (recall OPT = $\min_{\text{rank}-k\ P} |A(I - P)|_{1,2}$)

Problems with Additive Error

- Flattening
  - For a vector y, replace an entry y_i with t copies of y_i/t , preserve the 1-norm of y
  - Can form a new vector y' such that
    - y' has dimension at most $2n$
    - $\|y'\|_1 = \|y\|_1$
    - $\|y'\|_2 \leq O(n^{-\frac{1}{2}})\|y\|_1$

- Algorithm: efficiently compute a constant factor approximate solution P'
  - For every row a_i of A for which $|a_i(I-P')|_2 \geq \frac{2}{n} \cdot |A(I-P')|_{1,2}$, copy and rescale a_i
  - Let B be the new input matrix. Show $\|B(I-P')\|_F = O(\frac{1}{\sqrt{n}}) \|B(I-P')\|_{1,2}$
  - Additive error $\lambda^{1/2}\|X\|_{1,2} = \frac{1}{k^{1/2}} \|B-B_k\|_F \|X\|_{1,2} \leq \|B(I-P')\|_F \cdot n^{\frac{1}{2}} = O(OPT)$
  - Additive error is small enough!

Problems with Additive Error: p > 2

- For p > 2 our algorithm samples proportional to the p/2-powers of ridge leverage scores with no input flattening
- RLS is leverage score sampling on a concatenated matrix $[A; \lambda^{1/2} I]$
- Instead, leverage score sample from $[A; \lambda^{1/2} U]$ for a random orthonormal matrix U
- The scores are the same: $a_i^T \left(A^T A + \lambda^{\frac{1}{2}} U^T U \lambda^{\frac{1}{2}}\right)^{-1} a_i = a_i^T \left(A^T A + \lambda I\right)^{-1} a_i$ for any orthonormal matrix U
- Additive error in our embedding for a vector x now depends on $\lambda^{p/2} \|Ux\|_p^p$ and $|Ux|_p \ll |x|_2$ if U is random

Conclusion

- We resolve the dependence of k in the coreset size for ℓ_p subspace approximation
- Techniques use a combination of ridge leverage scores and flattening
- Seamlessly handle online/streaming settings
- Open directions
  - Coreset question: resolving the polynomial dependence on ε
  - Currently, coreset size is $k/\epsilon^{4/p}$ for p < 2 and $k^{\frac{p}{2}}/\epsilon^{p}$ for p > 2
- Relaxed notions:
  - if we only preserve the optimization solution, can we get fewer rows?
  - if we can output a slightly larger dimensional space, can we get fewer rows or efficient algorithms?